SLCCI (Sea Level Climate Change Initiative)

So here's the deal — this is part of a bigger ESA effort, the Climate Change Initiative, where they're trying to finally make proper use of thirty-plus years of Earth observation archives instead of letting all that satellite data just sit there. There are about eleven of these projects running in parallel, each one looking at a different climate variable, and ours is the sea level one. Basically the science side has already been doing the heavy lifting — building the algorithms, validating them, running a kind of prototype production setup — and now we're at the point where somebody needs to take that prototype and turn it into something that actually runs operationally, day in day out, and can be handed over to the people who need the data.

What I picture happening is a system that pulls in altimetry data from a bunch of different satellite missions, runs it through calibration and validation, cross-calibrates between missions so everything is consistent, and spits out two kinds of products: one is like a per-mission track of measurements, and the other is a merged, gridded product that blends all the missions together into one consistent climate record. On top of that there needs to be a proper archive, because reprocessing is going to happen again and again as algorithms improve or new calibration methods come out, so we can't afford to lose the baseline data or any of the intermediate stuff along the way.

The people using this thing day-to-day are going to be on two very different sides. Internally you've got the folks who actually run and maintain the production line — scientists who get pulled in when something needs deeper validation or when a new product has to be defined, people who just judge and sign off on product quality, people who handle the catalog and metadata, and then the more operational crowd who monitor the system, handle incidents, deal with user requests, that sort of thing. Externally, the audience is really the climate research community — people doing climate modelling work who just want reliable, well-documented sea level data they can trust and build on. We're not building this for the general public or commercial customers, it's a fairly specialized research crowd.

One thing I want to be upfront about: we don't need to worry about near-real-time delivery. There's a whole separate world of oceanographic services that care about getting data out within hours for things like operational forecasting, and that's just not us — we're purely about delayed-time and the long, reprocessed historical series for climate purposes, so anything geared at that fast-turnaround NRT use case can be dropped from consideration entirely. Similarly, all that formal IT service management process stuff, incident ticketing frameworks and so on, isn't something we need to bring in here, it just doesn't fit how this is going to be run.

Cost is a real concern too, this has to be something that's sustainable to operate long term, not just a one-off build. And it has to be flexible — new satellite missions are coming online

during the life of the project and the algorithms themselves are going to keep changing as the science matures, so whatever gets built can't be rigid, it needs to absorb new data sources and let people swap in improved processing logic without everything falling apart. There's also a push to not reinvent the wheel — there's existing infrastructure out there already doing something very similar for ocean data more broadly, and wherever it makes sense we'd rather lean on what's proven and working than build everything from a blank page. Quality and traceability of the data matter a lot too, since the whole point is feeding trustworthy numbers into the international climate change conversation, so whatever comes out the other end needs solid metadata and documentation attached, not just raw numbers.